

1. **Experiment:** Genomic DNA third-generation sequencing
2. **Time:** 2026.05.01-2026.05.21
3. **Member:** Zhangdong Liu, Yuxuan Wang, Kaili Jin, Yulongzi Piao, Yue Chao
4. **Method:**
 - (1) Cell collection:
 - ① Cells that had undergone dCas9-AID*Δ editing were harvested from culture flasks.
 - ② The cell suspension was centrifuged at $300 \times g$ for 5 min at room temperature.
 - ③ The supernatant was carefully removed, and the cell pellet was washed twice with ice-cold PBS.
 - ④ The cell pellet was resuspended in a small volume of PBS (e.g., 200 μ L) and transferred to a pre-labelled 1.5 mL microcentrifuge tube.
 - ⑤ The cells were centrifuged again at $300 \times g$ for 5 min at 4 $^{\circ}$ C, and the supernatant was completely removed.
 - (2) Sample preparation:
 - ① Genomic DNA was extracted from the cell pellet using a commercial genomic DNA extraction kit (TIANamp Genomic DNA Kit) according to the manufacturer's instructions.
 - ② The concentration and purity of the extracted genomic DNA were determined using a PacBio spectrophotometer. DNA samples with an A_{260}/A_{280} ratio between 1.8 and 2.0 were used for subsequent sequencing.
 - ③ The DNA sample was aliquoted into a pre-labelled 1.5 mL microcentrifuge tube and stored at -20 $^{\circ}$ C until shipment.
 - (3) Shipment and sequencing:
 - ① The prepared DNA samples were shipped to Tsingke Biotechnology on dry ice according to the company's sample submission guidelines.
 - ② Third-generation sequencing was performed by Tsingke Biotechnology, and the sequencing data were returned to us for further analysis.

5. Result:

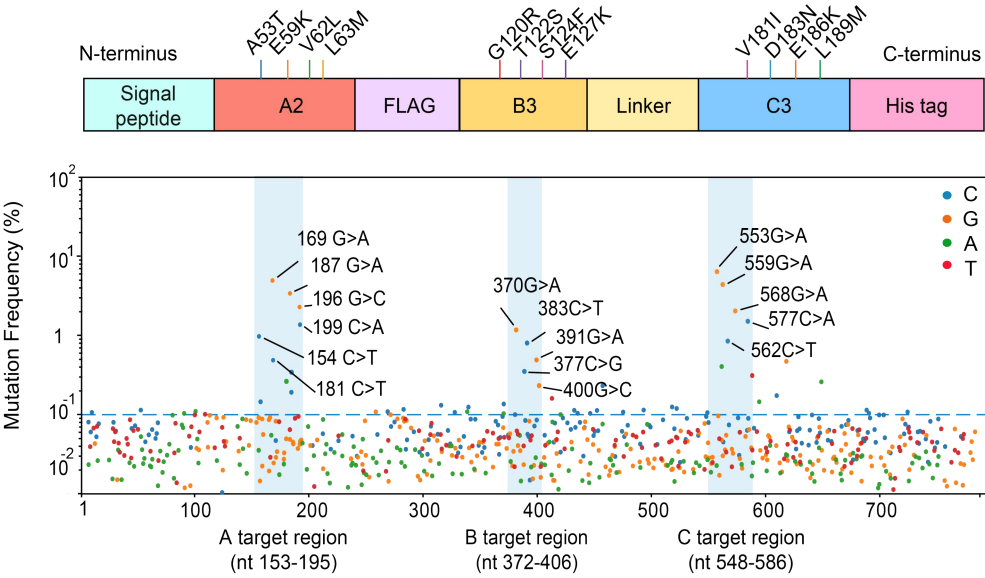


Figure 1. Representative high-frequency mutations within the A2, B3 and C3 regions. Per-nucleotide substitution frequencies relative to the 789-nt reference sequence. Position 1 of the reference sequence was defined as the origin of the x axis, and the start codon of the coding region was located at nt 13. The substitution frequency at each position was calculated as the number of high-quality reads containing a nucleotide substitution divided by the total number of reads covering that position. Colors indicate the corresponding

reference nucleotide, and the dashed line denotes the 0.1% frequency threshold used to define detectable mutated sites. The principal mutations in group A included G169>A (24.44%), G187>A (15.38%), G196>C (8.76%) and C199>A (4.99%); those in group B included G370>A (19.20%), C383>T (13.07%), G391>A (7.95%) and C377>G (5.67%); and those in group C included G553>A (24.68%), G559>A (16.98%), G568>A (9.21%) and C577>A (7.08%). All nucleotide positions are numbered according to the same 789-nt reference sequence.

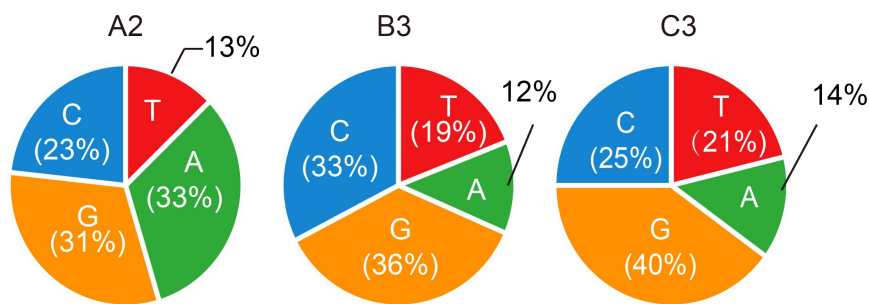


Figure 2. Composition of detectable mutated sites. Composition of detectable mutated sites in the A2-, B3- and C3-targeted groups, classified according to the reference nucleotide (A, T, C or G).